Rank-one Carlitz torsion has no degree-free bound

TLMC 00000000092 lizaixi01 24 October 2026

Precisely the printed parameter claim. The source asks for a torsion bound depending only on the rank r and q. We fix the coefficient ring $A = \mathbb{F}_2[T]$, its fraction field $K = \mathbb{F}_2(T)$, and rank $r = 1$. We exhibit finite extensions L_n/K and a rank-one Drinfeld A-module over each L_n with at least 2^n distinct L_n-rational torsion points. Thus no bound depending only on these fixed r, q can work uniformly over all finite extensions. Here $q = 2$ is the size of the coefficient field of A. We do not assert that the full constant field of every L_n has been computed.

This is not a counterexample to the standard degree-bounded formulation. Poonen's Conjecture 2 fixes A, r, d and restricts $[L : K] \le d$; Conjecture 1 fixes L itself. The following paragraph explicitly explains why allowing arbitrary degrees prevents a uniform bound [1]. Neither the fixed-d nor the fixed-L conjecture is refuted here.

The actual Drinfeld action. Let $\theta = T \in K$. Over a field extension L/K, define

$$\phi_T(x) = x^2 + \theta x, \qquad P_0(X) = X, \qquad P_{n+1}(X) = P_n(X)^2 + \theta P_n(X). \tag{1}$$

For $a(T) = \sum_i a_i T^i \in A$, set

$$\phi_a(X) = \sum_i a_i P_i(X). \tag{2}$$

All coefficients and all terms of the parameter polynomial are used. The map $x \mapsto x^2 + \theta x$ is an actual $\mathbb{F}_2$-linear endomorphism. Evaluating a in its endomorphism ring gives a genuine A-action. The proof verifies that (2) evaluates to exactly that action. Consequently it proves, as polynomial identities,

$$\phi_0 = 0, \quad \phi_1 = X, \quad \phi_{a+b} = \phi_a + \phi_b, \quad \phi_{ab} = \phi_a \circ \phi_b, \quad \phi_c = cX \ (c \in \mathbb{F}_2).$$

Each P_i, and hence each ϕ_a, belongs to the actual span of $X, X^2, X^4, \ldots$. The proof establishes this by closure under squaring and linear combinations; these are precisely the $\mathbb{F}_2$-linear polynomials in the concrete Drinfeld definition [2].

Differentiating (1)–(2) gives

$$P_n'(X) = \theta^n, \qquad \phi_a'(X) = a(\theta). \tag{3}$$

The map $a \mapsto a(\theta)$ is the actual inclusion $A \hookrightarrow L$. Thus the action is faithful and has generic A-characteristic. Moreover $\phi_T = X^2 + \theta X$ has highest Frobenius exponent one: its rank is one. The formal rank is computed from the actual degree as $\log_2(\deg_X \phi_T) = 1$, not assigned as a constant. This construction agrees with the usual Carlitz recursion [3].

Actual finite extensions and exact division-point counts. For each integer $n \geq 0$, take L_n to be the native splitting field over K of the specific polynomial P_n. It is a finite-dimensional field extension of K; no algebraic closure is substituted for L_n. Induction using (1) proves

$$\deg_X P_n = 2^n. \tag{4}$$

Indeed the square term has degree 2^{n+1} and the remaining term has degree at most 2^n, so there is no leading-term cancellation. Since $\theta \neq 0$, (3) gives a nonzero constant derivative. An explicit Bezout identity with that constant proves separability. Splitting and separability therefore give the exact count

$$D_n := \{x \in L_n : P_n(x) = 0\}, \qquad |D_n| = 2^n. \tag{5}$$

These are actual roots and their actual cardinality, not a stipulated count.

The points lie in the genuine torsion submodule. In the native A-module induced by ϕ_T, the proof establishes

$$a \cdot x = \phi_a(x), \qquad x \in \operatorname{Tor}_A(\phi, L_n) \iff \exists a \in A \setminus \{0\} : \phi_a(x) = 0.$$

It also proves that the recursive polynomial commutes with scalar extension from K to L_n. Thus $\phi_{T^n} = P_n$ over this very field. Every element of D_n is killed by the actual nonzero polynomial T^n. The proof constructs an injective map

$$D_n \hookrightarrow \operatorname{Tor}_A(\phi, L_n), \qquad 2^n \leq \#\operatorname{Tor}_A(\phi, L_n). \tag{6}$$

The right side uses cardinal numbers for the whole native torsion submodule. It never uses a natural-cardinality convention that returns zero on an infinite type, and does not need to compute all torsion points.

No bound with only the two printed parameters. Suppose a finite bound B worked for all the above fields and modules. Taking $n = B$ in (6) yields $2^B \leq B$, whereas $B < 2^B$ for every natural number B. This contradiction covers every finite bound while $r = 1$, $q = 2$ and even the coefficient ring A remain fixed. Only the finite extension and its permitted degree vary.

Formal and interpretive limits. Lean constructs the real rational-function field, splitting fields, finite-dimensional extensions, $\mathbb{F}_2$-linear endomorphism action, additive-polynomial identities, generic-characteristic inclusion, rank computation, exact root sets and native torsion submodules. The explicit algebraic Drinfeld structure records the standard polynomial addition/composition and tangent-action conditions; it is not an axiom asserting that arbitrary modules are Drinfeld modules. The project pins Lean 4.33.0 and Mathlib `db584cd6d46c92f209a44c0f1c829460d327499d`. Its evidence records source-bound compilation, all-namespace ordinary `#print axioms` and the official fresh-kernel replay.

No admitted proofs or extra axioms are used. The proof does not use `native_decide`. Production timings are not controlled efficiency measurements.

References

[1] B. Poonen, *Torsion in rank 1 Drinfeld modules and the uniform boundedness conjecture*, arXiv:math/9507217, v1 (1995), printed p. 2 (torsion and the fixed-degree formulations).

[2] B. Poonen, *Introduction to Drinfeld modules*, notes dated 29 December 2021, printed pp. 4–5, Definition 3.2 and Example 3.4, author's PDF.

[3] K. Conrad, *Carlitz extensions*, author's notes, printed pp. 1–3 and 7 (recursion, separability and torsion).